Routing Table

- Static routing table: created manually
- Dynamic routing table: updated periodically by using one of the dynamic routing protocols such as RIP, OSPF, or BGP.

Routing Protocols

- A router consults a routing table when a packet is ready to be forwarded
- The routing table specifies the optimum path for the packet: static or dynamic
- Internet needs dynamic routing tables to be updated as soon as there is a change
- There are two routing: Unicast routing and multicasting routing
- RIP (Routing Information Protocol), OSPF (Open Shortest Path First), BGP (Border Gateway Protocol)

Optimization

- Which of the available pathways is the optimum pathway ?
- One approach is to assign a cost for passing through a network, called metric
- Total metric is equal to the sum of the metrics of networks that comprise the route

Router chooses the route with shortest (smallest) metric

RIP (Routing Information Protocol):

- The cost of passing each network is one hop count.
- If a packet passes through 10 networks to reach the destination, the total cost is 10 hop counts.

OSPF (Open Shortest Path First):

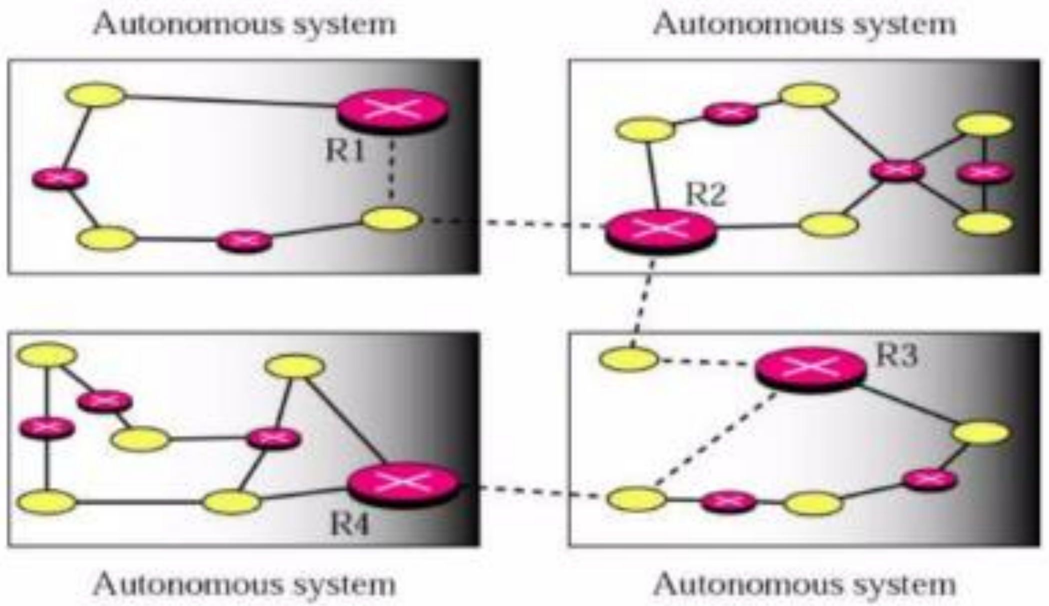
- Administrator can assign cost for passing a network based on type of service required.
- OSPF allows each router to have more than one routing table based on required type of service. Such as Maximum throughput or minimum delay.

BGP (Border Gateway Protocol):

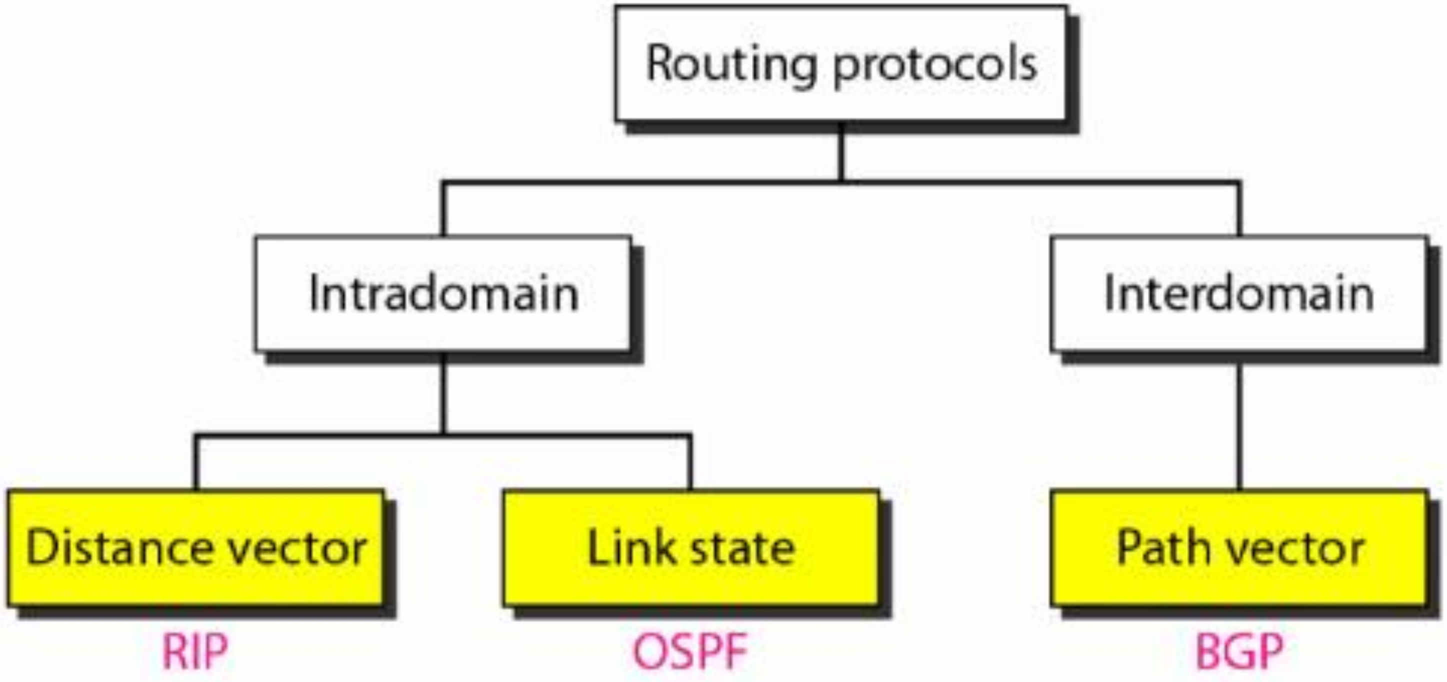
- Criterion is the policy, which is set by the administrator(speaker node).

Intra- and Interdomain Routing

- AS (autonomous system): A group of networks and routers under the authority of a single administration
- Intradomain routing: inside an AS
- Interdomain routing: between ASs
- R1, R2, R3, and R4 use a intradomain and an interdomain routing protocol.
- The other routes use only intradomain routing protocols



Popular (Unicast) Routing Protocols



Routing Algorithm classification

1. Distance vector algorithms: router knows physically-connected neighbors, link costs to neighbors, iterative process of computation, exchange of partial information with neighbors.
2. Link state algorithms: all routers have complete topology, link cost information.

Metric of different routing protocols

- Metric is the cost assigned for passing through a network.
- The total metric of a particular route is equal to the sum of the metrics of networks that comprise the route.
- A router chooses the route with smallest metric.

RIP (Routing Information Protocol):

- The cost of passing each network is one hop count.
- If a packet passes through 10 networks to reach the destination, the total cost is 10 hop counts.

OSPF (Open Shortest Path First):

- Administrator can assign cost for passing a network based on type of service required.
- OSPF allows each router to have more than one routing table based on required type of service. Such as Maximum throughput or minimum delay.

BGP (Border Gateway Protocol):

- Criterion is the policy, which is set by the administrator(speaker node).

DISTANCE VECTOR ROUTING

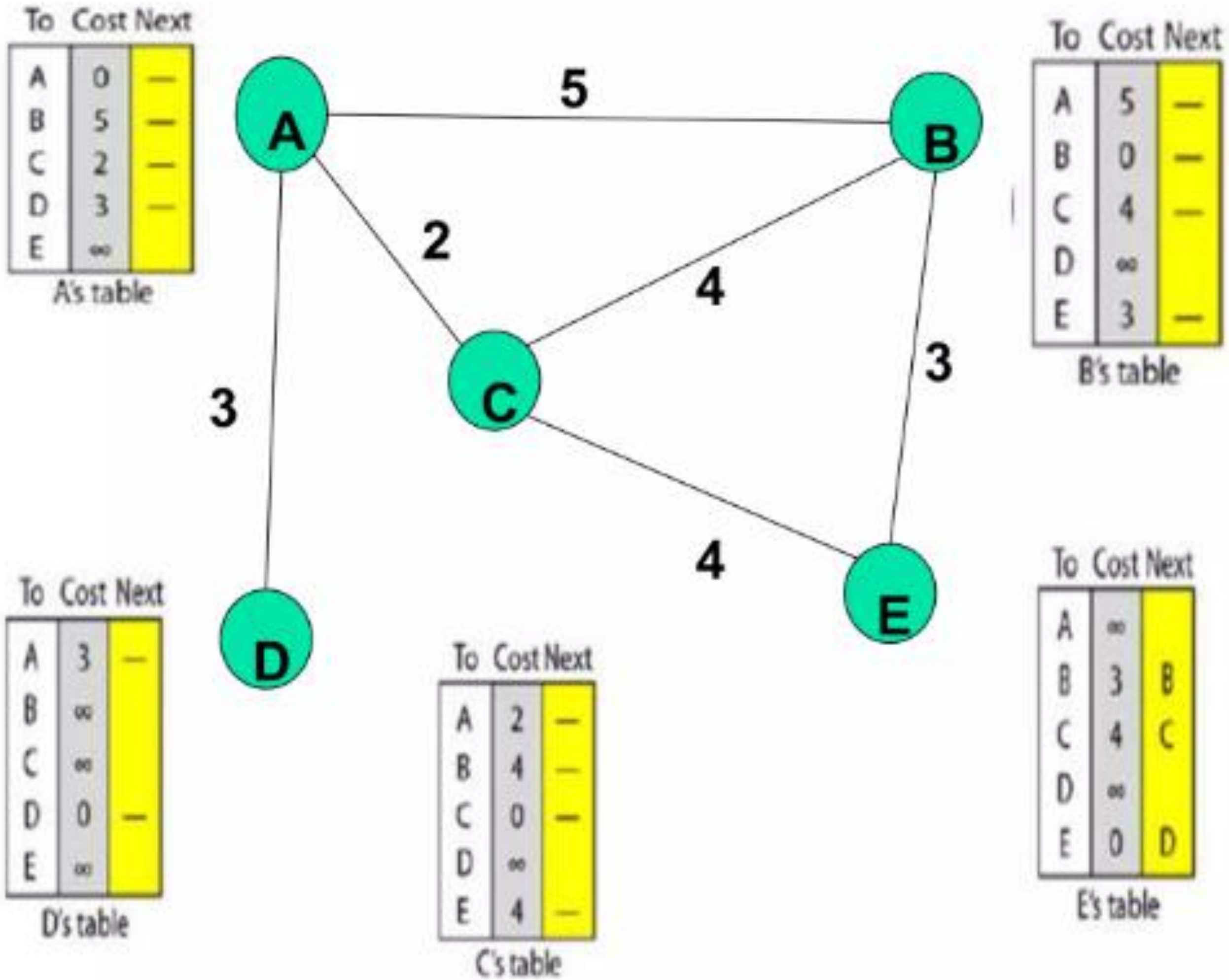
- Each node maintains a set of triples
  (Destination, Cost, NextHop)
- Node knows the cost to each neighbor.
- Directly connected neighbors exchange updates
  - Periodically (normally every 30 secs)
  - whenever table changes (called *triggered* update)
- Each update is a list of pairs:
  (Destination, Cost)

Routing Table

- Neighbors exchange table entries
- Determine current best next hop
- For each destination list:
  - Next Node
  - Distance
- **“In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change.”**
- The least-cost route between any two nodes is the route with minimum distance.
- Each node maintains a vector(table) of minimum distances to every node

Initialization of tables in DV Routing

- At the beginning, each node can know only the distance between itself and its immediate neighbors



Distance Vector Routing: Sharing

- In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change

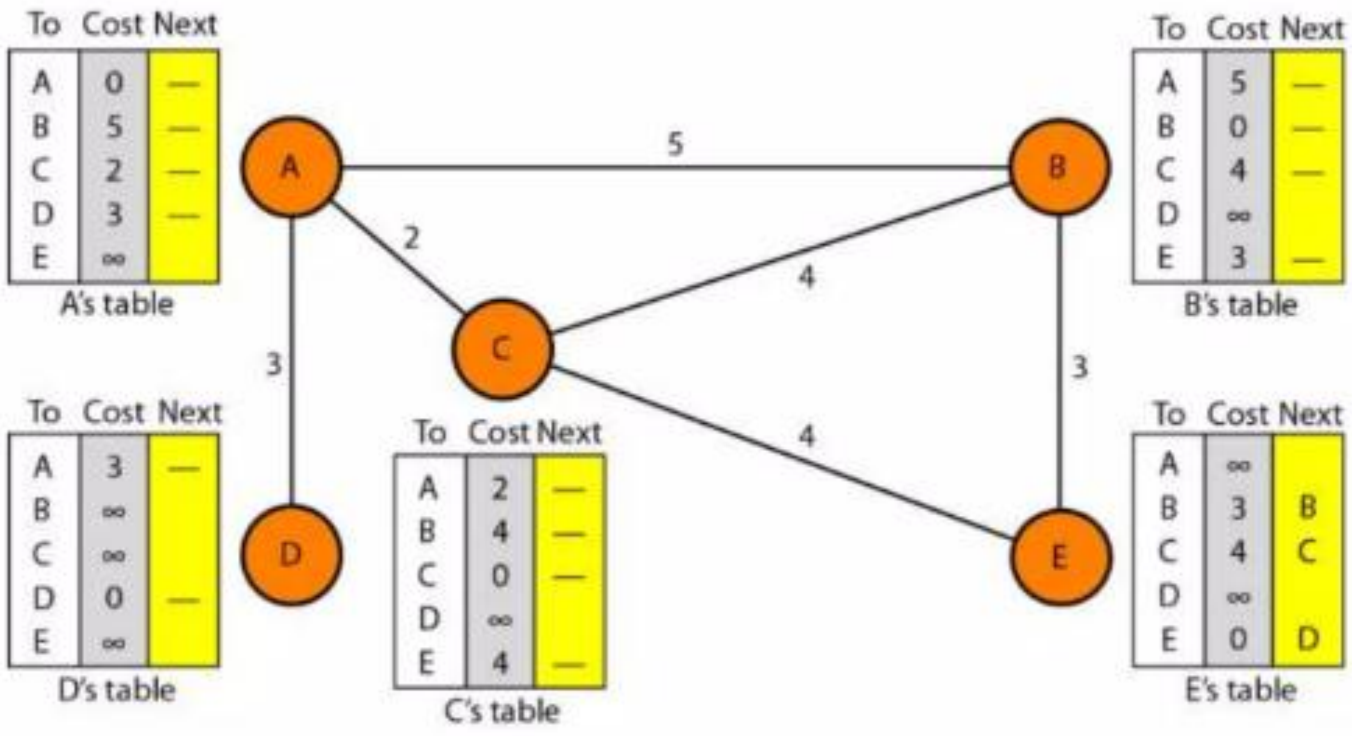
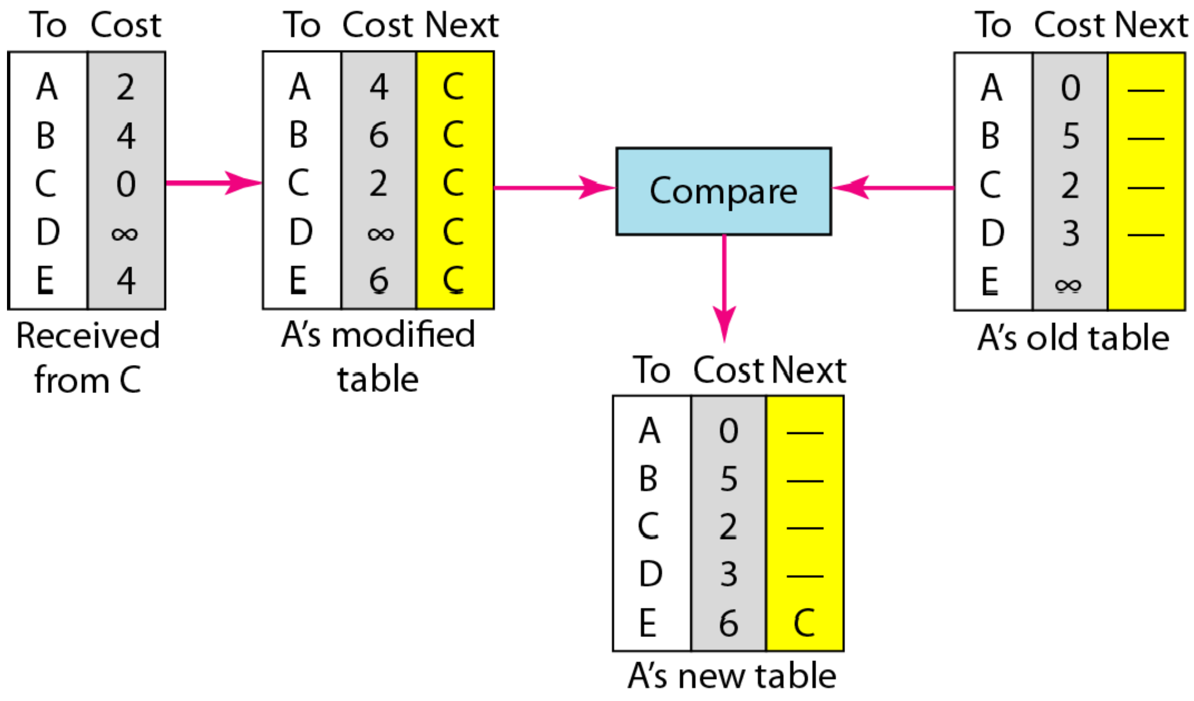


Figure 22.16 *Updating in distance vector routing*



Updating in Distance Vector Routing

When a node receives a two-column table from a neighbor, it need to update its routing table
Updating rule:

Choose the smaller cost. If the same, keep the old one
If the next-node entry is the same, the receiving node chooses the new row

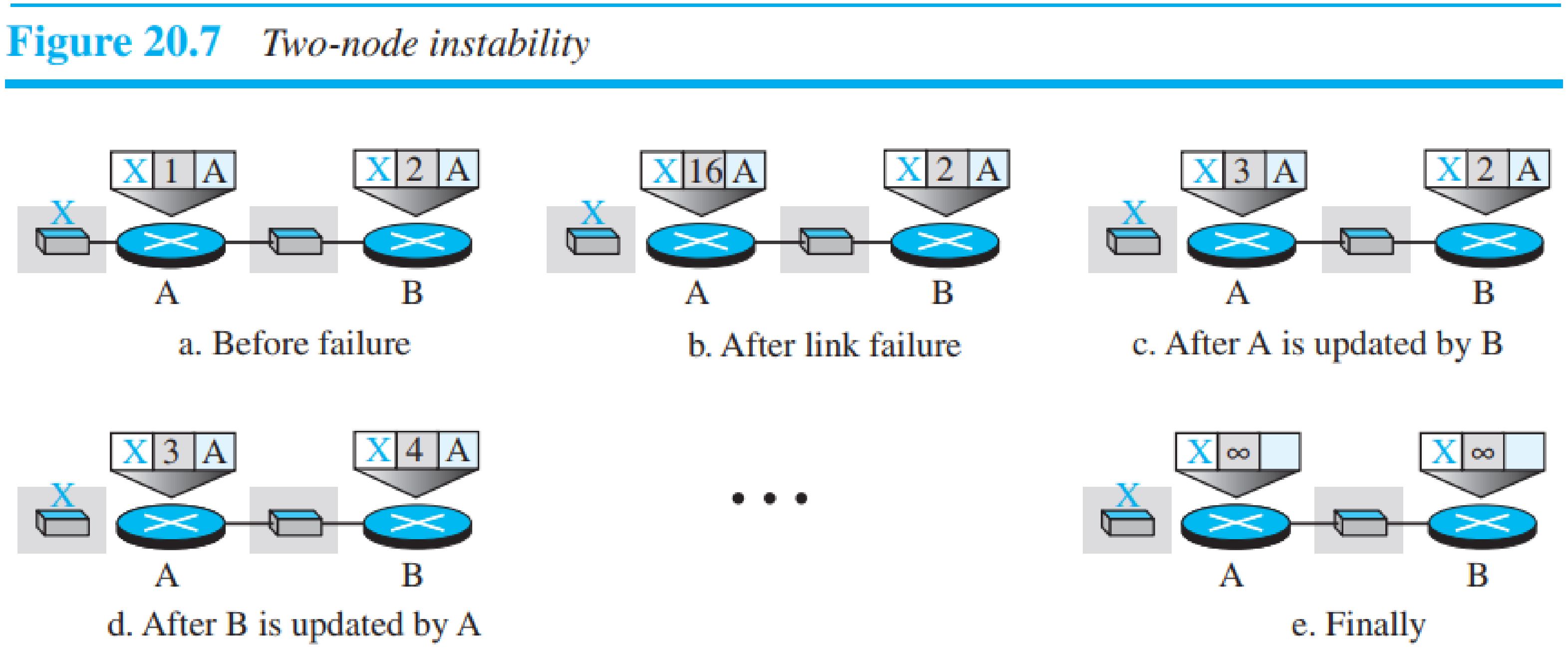
When to Share

- Periodic update: A node sends its routing table, normally every 30 s
- Triggered update: Anode sends its two-column routing table to its neighbors anytime there is a change in its routing table

Count to Infinity

- A problem with distance vector routing is that any decrease in cost (good news) propagates quickly, but any increase in cost (bad news) propagates slowly.
- For a routing protocol to work properly, if a link is broken (cost becomes infinity), every other router should be aware of it immediately, but in distance vector routing, this takes some time. The problem is referred to as count to infinity.
- Solution can be given through **Split Horizon** and **Poison Reverse**

Figure 22.17 *Two-node instability*



Split Horizon

- One solution to Count to Infinity problem is called Split Horizon
- In this strategy, instead of flooding the table through each interface, each node sends only part of its table through each interface
- Node B thinks that the optimum route to reach X is via A, it does not need to advertise this piece of information to A
- Taking information from node A, modifying it, and sending it back to node A is what creates the confusion.
- In our scenario, node B eliminates the last line of its forwarding table before it sends it to A. In this case, node A keeps the value of infinity as the distance to X.
- Later, when node A sends its forwarding table to B, node B also corrects its forwarding table. The system becomes stable after the first update: both node A and node B know that X is not reachable.

Poison Reverse

- Using the split-horizon strategy has one drawback.
- Normally, the corresponding protocol uses a timer, and if there is no news about a route, the node deletes the route from its table.
- When node B in the previous scenario eliminates the route to X from its advertisement to A, node A cannot guess whether this is due to the split-horizon strategy (the source of information was A) or because B has not received any news about X recently.
- In the poison reverse strategy B can still advertise the value for X, but if the source of information is A, it can replace the distance with infinity as a warning: "Do not use this value; what I know about this route comes from you."

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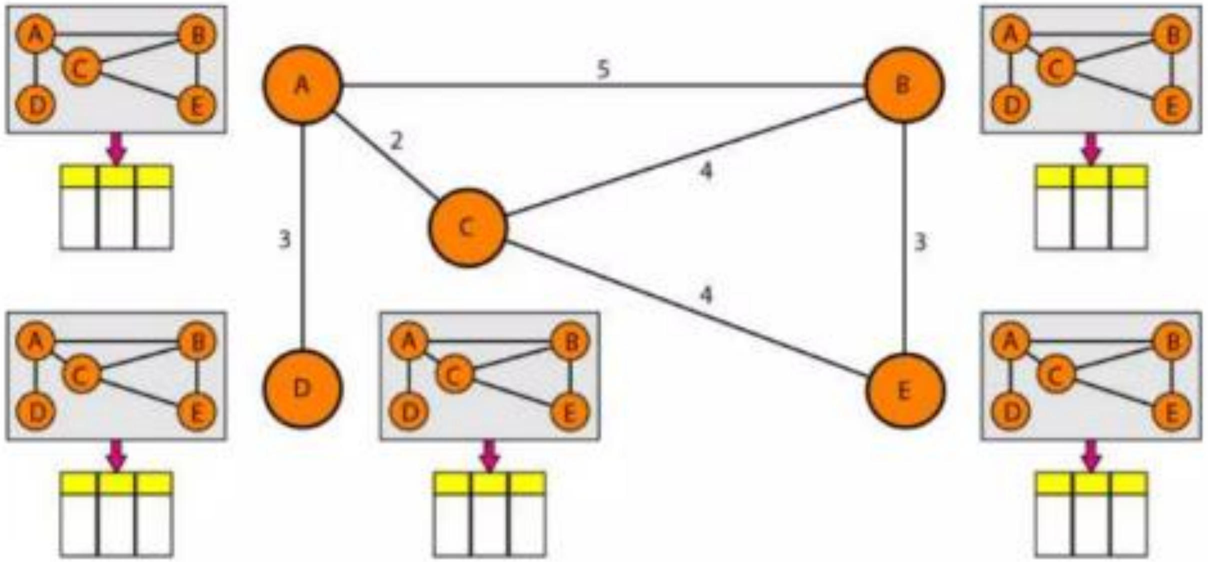
Link State Routing

- Each node has the entire topology of the domain- the list of nodes and links, how they are connected including type, cost, and condition of the links(up or down)
- Node can use **Dijkstra's algorithm** to build a routing table



Link State Knowledge

- Each node has partial knowledge: it know the state (type, condition, and cost) of its links. The whole topology can be compiled from the partial knowledge of each node

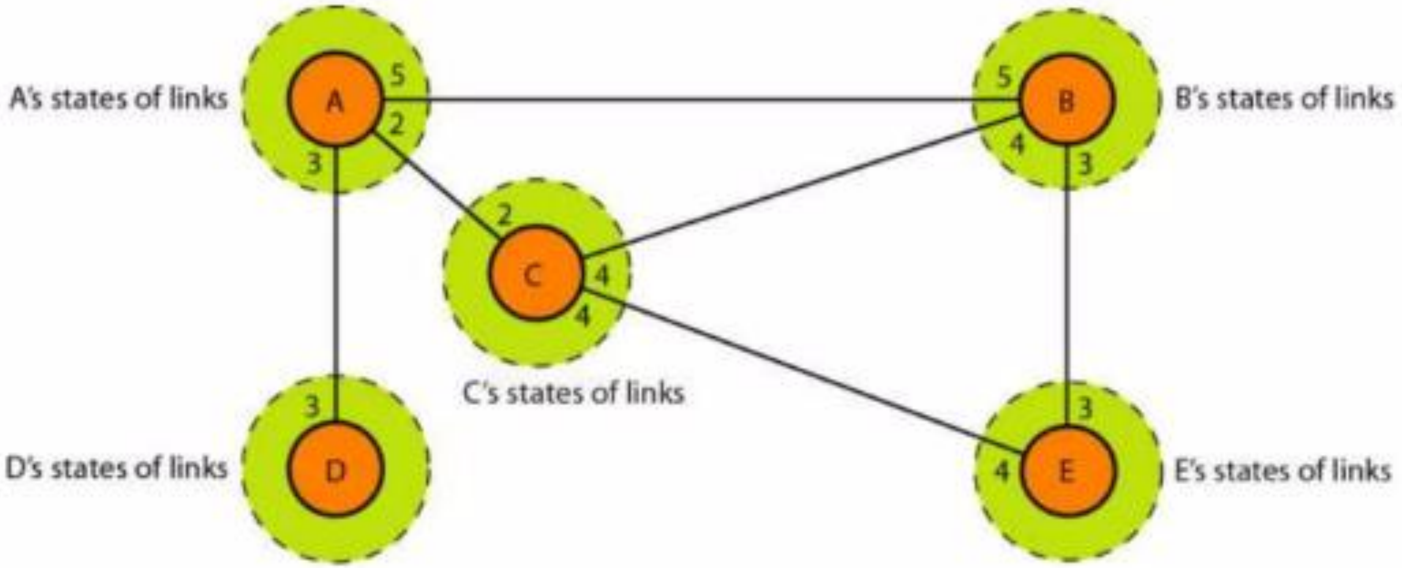
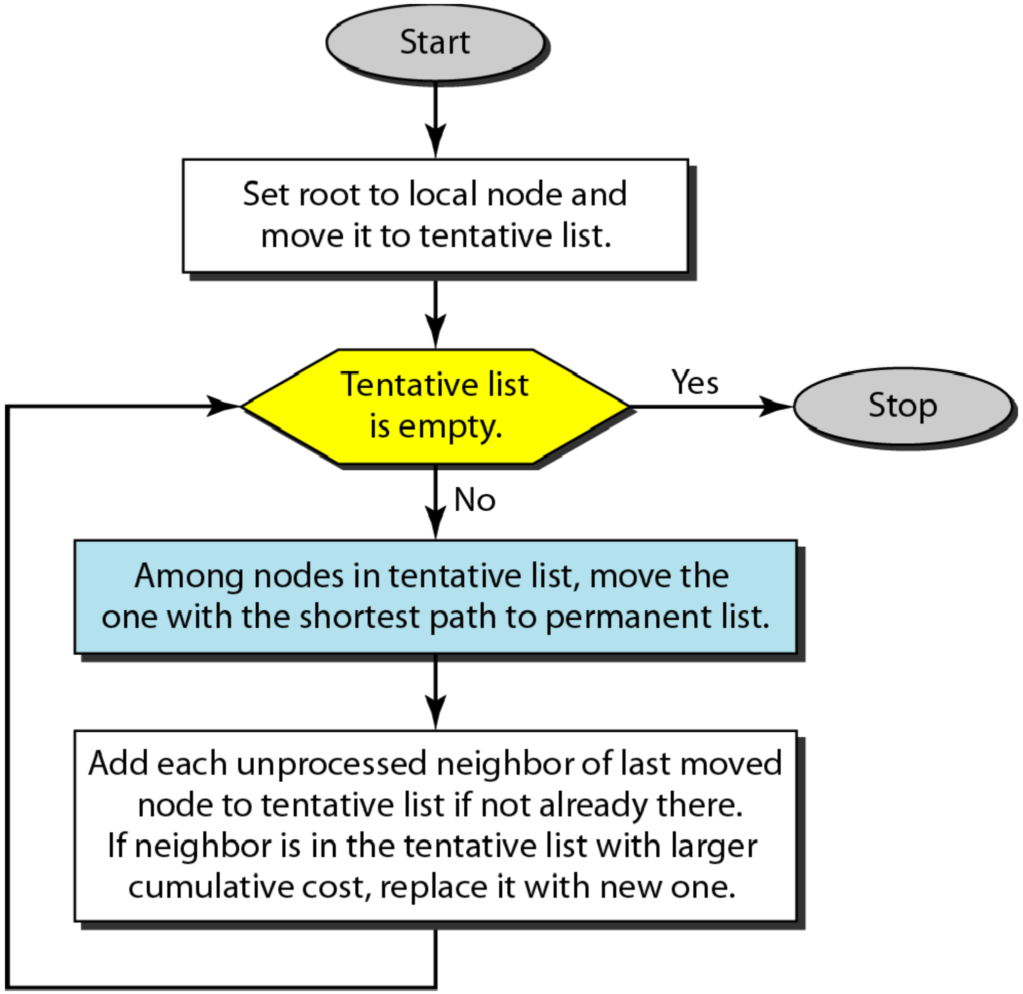
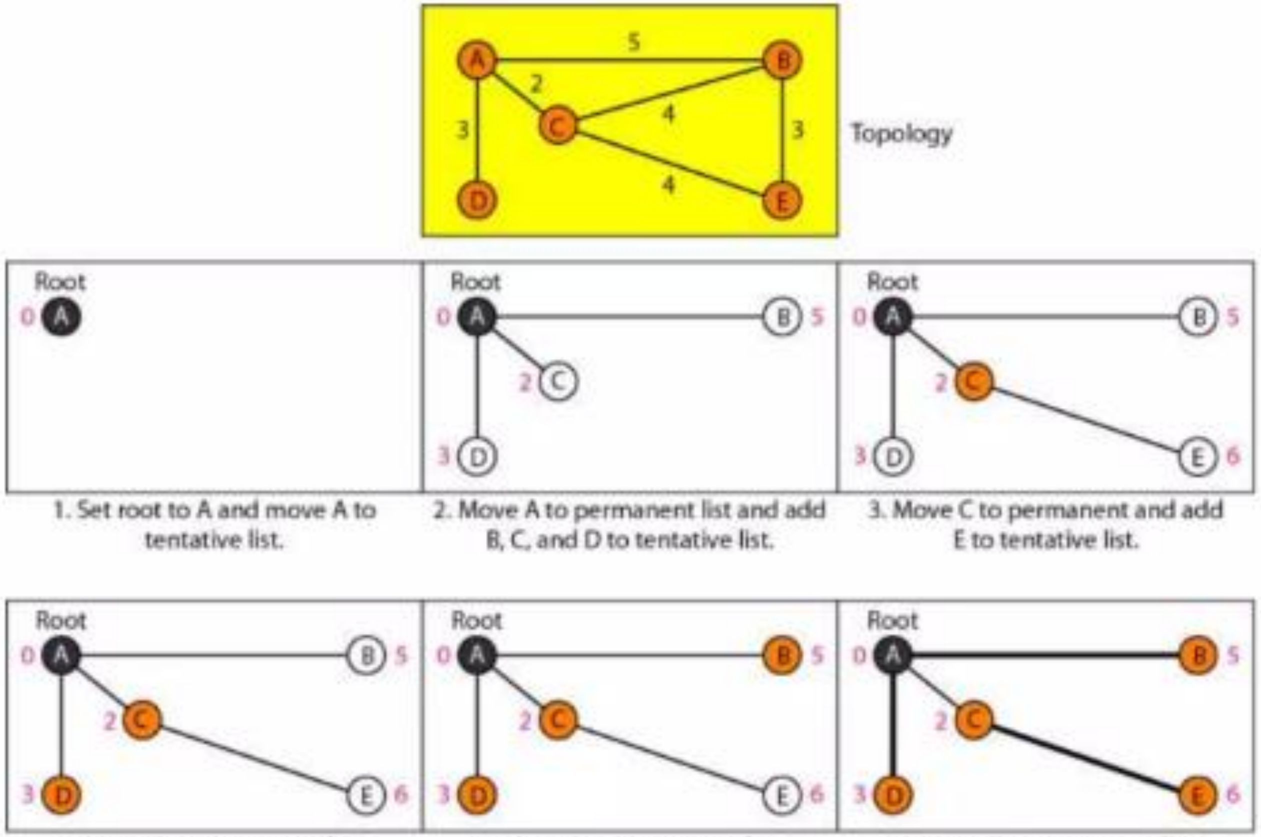


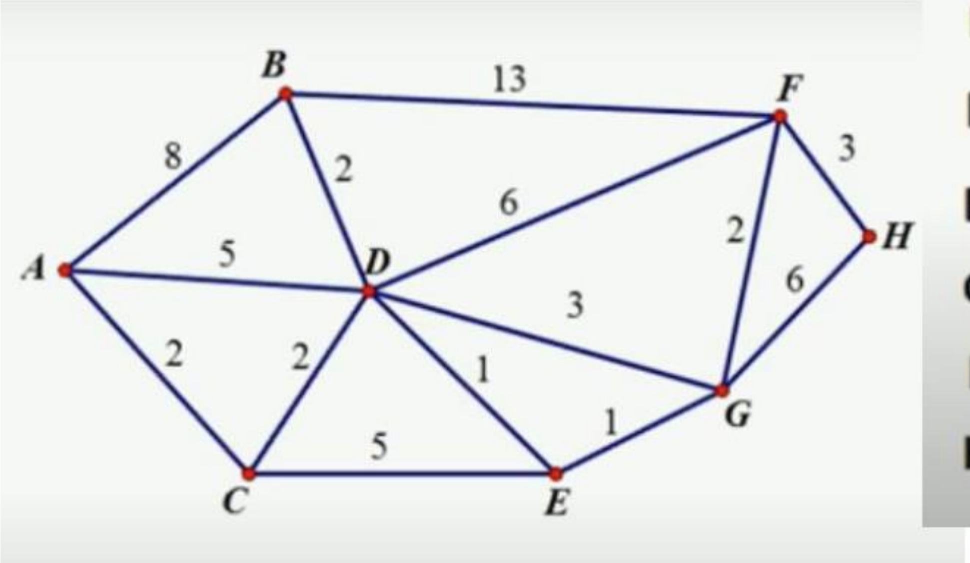
Figure 22.22 *Dijkstra algorithm*



Example of Dijkstra Algorithm



1. Set root to A and move A to tentative list.
2. Move A to permanent list and add B, C, and D to tentative list.
3. Move C to permanent and add E to tentative list.
4. Move D to permanent list.
5. Move B to permanent list.
6. Move E to permanent list (tentative list is empty).



V	A	B	C	D	E	F	G	H
A	$\boxed{0_A}$	8_A	2_A	5_A	∞	∞	∞	∞
C		8_A	$\boxed{2_A}$	4_C	7_C	∞	∞	∞
D		6_D		$\boxed{4_C}$	5_D	10_D	7_D	∞
E		6_D			$\boxed{5_D}$	10_D	6_E	∞
B		$\boxed{6_D}$				10_D	6_E	∞
G						8_G	$\boxed{6_E}$	12_G
F						$\boxed{8_G}$		11_F
H								$\boxed{11_F}$